

Western Ghats Deforestation & Land Cover Change Detection

Region: Coorg | Years: 2020 → 2024

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Alert Level: Low Risk

Forest Cover Summary

Metric	Value
Forest Cover (Year 1)	54.33%
Forest Cover (Year 2)	57.47%
Forest Loss	0.00%
Forest Gain	3.14%
Net Forest Change	+3.14%

Land Cover Percentages

Class	Year 2020 %	Year 2024 %	Change %
Forest	54.33%	57.47%	+3.14%
Agriculture	0.02%	0.00%	-0.02%
Urban	21.60%	18.25%	-3.35%
Water	14.75%	10.04%	-4.71%
Barren	9.31%	14.25%	+4.94%

Key Land Cover Transitions

Transition	Area (%)
Forest → Urban	2.359%
Forest → Agriculture	0.000%
Forest → Barren	0.095%
Agriculture → Urban	0.003%

Recommendations

- Urban expansion detected near forest boundary.

Academic Analysis & Ecological Justification

1. Reasoning for Land Cover Shifts

A positive trend with a 3.14% increase in forest cover suggests effective local conservation efforts and natural regeneration, aligning with recent afforestation drives under the Green India Mission. A concerning reduction of 4.71% in water bodies correlates with reduced groundwater recharge due to catchment degradation, exacerbating hydrological vulnerability in the region as noted by the Central Water Commission. The 4.94% increase in barren land highlights soil erosion and localized resource extraction (such as quarrying), which strip the land of its natural ecological resilience.

2. Steps for Ecological Balance

- **Enforcement of Eco-Sensitive Zones (ESZs):** Strict implementation of zoning laws as recommended by the Kasturirangan Committee to restrict hazardous industrial and mining activities in vulnerable tracts.
- **Agroforestry Integration:** Transitioning from monoculture plantations to biodiversity-friendly agroforestry systems in buffer areas to restore soil fertility and create wildlife corridors.
- **Hydrological Restoration:** Rejuvenating degraded catchments through watershed management and afforestation to secure the water table and sustain regional hydrology.

3. Conclusive Points

- **Habitat Fragmentation & Edge Effects:** The observed transitions in Coorg reveal a concerning pattern. The conversion of natural habitats to anthropogenic land uses isolates endemic species and diminishes overall biodiversity capacity.
- **Ecological Vulnerability:** The land-cover dynamics underscore a precarious balance. Without proactive management, even minor anthropogenic perturbations could trigger irreversible ecological decline in this global biodiversity hotspot.
- **Imperative for Data-Driven Policy:** Real-time, satellite-derived insights emphasize the critical need for executing ground-level environmental regulations promptly. Continued unmonitored land-cover alterations will irreversibly compromise the long-term ecological and economic stability of peninsular India.

Visual Maps

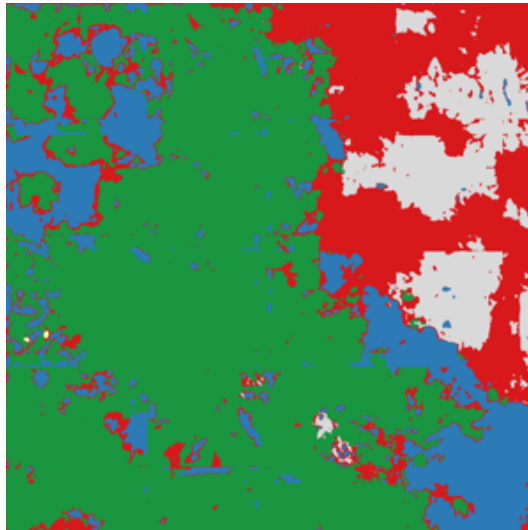
Satellite Image — Year 2020



Satellite Image — Year 2024



Segmentation Map — Year 2020



Segmentation Map — Year 2024



Change Detection Map (Red=Forest Loss, Green=Forest Gain, Orange=Urbanisation)

